AI-Powered Telecom Network Operations Assistant

System architecture · local execution · Python 3.12 · every LLM step has a deterministic fallback

Offline build step · Deterministic · no LLM · LLM component · Storage / artifact · Tool call / data read · Feedback loop

01 OFFLINE BUILD

`python -m scripts.bootstrap` `no LLM`

Dataset
5G NetOps · Hugging Face
100k rows · 27 KPIs

Preprocessing
Normalize · p75/p90 baselines
19 symptoms → 8 categories · P1–P4

Chunking
Case narratives + runbook sections
1 chunk / case · 1 chunk / section

Embeddings
BAAI/bge-small-en-v1.5
384-d · CPU

Vector Database
ChromaDB · persistent, local
100k cases · 81 KB chunks

Models + Baselines
joblib models · baselines.json
processed cases.parquet

ML Training
Fault-risk classifier
Throughput regressor

KB Runbooks
9 Markdown runbooks
3GPP / ITU-T refs

BM25 Index
Keyword index over
case narratives

dense + keyword indexes

models + baselines

02 ONLINE REQUEST

`uvicorn src.main:app` `npm run dev · frontend-react`

Retriever
Hybrid dense + BM25
Reciprocal-rank fusion
Rerank with feedback prior
k = 60 · metadata filters

LLM / RAG Pipeline
Evidence packs, citable IDs
JSON repair · cache · breaker
run budget · daily cap

LLM Gateway
External model provider

Agents LangGraph state graph · deterministic fallback per node

revise · max 1

Classifier category · scope

Analyst root causes

Recommender ordered actions

Judge pass · revise · fail

DETERMINISTIC TOOLS · NO LLM

KPI analyzer bands · anomalies

Segment statistics co-occurrence lifts

ML predictors fault risk · throughput

Similar-case search via the Retriever

Prioritization P1–P4 · impact

Guardrails injection · scope · PII

Fallbacks: rules classifier · runbook causes & actions · rule checks
Reduced mode (no key · breaker open · cap hit): rules → retrieve → answer

API Layer
FastAPI /api/v1
13 endpoints
Guardrails first

User Interface
React · NetOps Triage
Investigate · Cases · Simulate · Quality

Beta-smoothed prior → rerank

tool call · similar cases / runbook sections

SQLite
artifacts/

`app.db` runs · ratings · votes

`llm_cache.db` validated replies

`llm_spend.db` daily spend cap

feedback · ratings + relevance votes